DATABASE DESIGN DOCUMENT

Predictive Inventory Management and Restock Alert System

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1. Overview

This document defines the relational database schema for the Predictive Inventory Management and Restock Alert System.

The database is designed to:

- Track finished products and raw materials
- Record sales transactions
- Store demand forecast results
- Generate and manage restock alerts
- Support backend automation and forecasting integration

The schema follows normalization principles (up to 3NF) to reduce redundancy and ensure data integrity.

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2. Entity Definitions

2.1 Product

Description:
Stores information about finished goods available for sale.

Attributes:

Attribute	Data Type	Description
product_id	INT (PK)	Unique product identifier
name	VARCHAR(100)	Product name
volume_ml	INT	Product volume in milliliters
unit_price	DECIMAL(10,2)	Selling price per unit

Attribute	Data Type	Description
current_stock	INT	Current stock level
stock_quantity	INT	Safety stock
total_value	DECIMAL(10,2)	Total stock value
reorder_level	INT	Minimum threshold before restocking
expiry_date	DATE	Expiry date of product batch
created_at	TIMESTAMP	Record creation timestamp

Primary Key:
product_id

2.2 Sales

Description:
Stores transactional sales data for products. Used for demand forecasting and inventory updates.

Attributes:

Attribute	Data Type	Description
sale_id	INT (PK)	Unique sale identifier
product_id	INT (FK)	References Product
sale_date	DATE	Date of sale
quantity_sold	INT	Units sold
total_amount	DECIMAL(10,2)	Total sale value
created_at	TIMESTAMP	Record creation timestamp

Primary Key:
sale_id

Foreign Key:
product_id → Product(product_id)

2.3 DemandForecast

Description:
Stores predicted demand results generated by the forecasting model.

Attributes:

Attribute	**Data Type**	**Description**
forecast_id	INT (PK)	Unique forecast identifier
product_id	INT (FK)	References Product
forecast_period	VARCHAR(50)	Forecast interval (e.g., Weekly, Monthly)
predicted_demand	INT	Forecasted quantity
created_on	DATE	Date forecast generated
created_at	TIMESTAMP	Record creation timestamp

Primary Key:
forecast_id

Foreign Key:
product_id → Product(product_id)

2.4 RawMaterial

Description:
Stores inventory information for production raw materials.

Attributes:

Attribute	**Data Type**	**Description**
raw_material_id	INT (PK)	Unique raw material identifier
reorder_id	VARCHAR(20)	Reorder tracking reference
item_name	VARCHAR(100)	Name of raw material

Attribute	Data Type	Description
item_description	VARCHAR(200)	Description
stock_location	VARCHAR(50)	Storage location
unit	VARCHAR(50)	Measurement unit (litres, kg, pcs)
quantity_in_stock	DECIMAL(10,2)	Available quantity
reorder_level	DECIMAL(10,2)	Minimum threshold
unit_cost	DECIMAL(10,2)	Cost per unit
expiry_date	DATE	Expiry date
created_at	TIMESTAMP	Record creation timestamp

Primary Key:
raw_material_id

2.5 RestockAlert

Description:
Stores restocking alerts for both products and raw materials.

Attributes:

Attribute	Data Type	Description
alert_id	INT (PK)	Unique alert identifier
product_id	INT (FK, nullable)	References Product
raw_material_id	INT (FK, nullable)	References RawMaterial
alert_type	ENUM	Type of alert (Product, RawMaterial)
reorder_point	DECIMAL(10,2)	Trigger threshold
alert_date	DATE	Date alert generated
status	ENUM	Alert status (Pending, Ordered, Resolved)

Attribute	Data Type	Description
created_at	TIMESTAMP	Record creation timestamp

Primary Key:
alert_id

Foreign Keys:

- product_id → Product(product_id)
- raw_material_id → RawMaterial(raw_material_id)

Note: Only one of product_id or raw_material_id will be populated per alert.

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3. Entity Relationships

Relationship	Type	Description
Product → Sales	1 : Many	One product can have multiple sales records
Product → DemandForecast	1 : Many	One product can have multiple forecasts
Product → RestockAlert	1 : Many	One product can generate multiple alerts
RawMaterial → RestockAlert	1 : Many	One raw material can generate multiple alerts

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4. Design Rationale

1. Separation of transactional, predictive, and inventory data ensures scalability.
2. DemandForecast is isolated from Sales to prevent mixing historical and predictive data.
3. RestockAlert centralizes alert management for both products and raw materials.
4. Nullable foreign keys in RestockAlert allow a unified alert tracking system.
5. Schema supports future automation logic and ML integration.